Practical - 14

Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition: Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

A. Theory

1. What is an admin dashboard?

Answer: It is a central page that summarizes important system information and gives authorized administrators access to management tools.

2. What is role-based access control?

Answer: RBAC grants permissions according to roles. The server must check the role on every protected page and action.

3. Why load statistics dynamically?

Answer: Database-backed counts and recent records keep the dashboard current without manually editing page content.

4. What is an audit log?

Answer: An audit log records important actions, including who performed an action, what happened, which record was affected, and when it occurred.

B. Implementation / Procedure

1. Protect dashboard

Start the session and reject users who are not authenticated administrators before outputting dashboard data.

2. Query statistics

Use SQL COUNT queries for users, students, and events. Fetch the latest registrations with ORDER BY created_at DESC and a suitable LIMIT.

3. Dynamic menu

Build navigation based on the verified role and ensure each destination independently checks permissions.

4. Audit log schema

Create audit_logs with id, user_id, action, entity_type, entity_id, details, and created_at fields.

5. Record actions

After a successful administrative operation, insert an audit entry using a prepared statement. Avoid logging passwords, tokens, or other secrets.

6. Display dashboard

Use summary cards and a recent-registration table. Escape database content before rendering and show sensible empty/error states.

C. Test Cases

Test Case	Input / Action	Expected Result
Admin access	Log in as admin	Dashboard and admin menu appear
Student access	Log in as student and open admin URL	Access is denied or redirected
Dynamic counts	Add a student/event	Corresponding count changes
Audit logging	Perform an admin update	A log entry records the action and actor
Menu visibility	Compare admin and student roles	Only role-appropriate links are displayed

D. Key Questions / Analysis / Interpretation

1. Are dashboard values loaded dynamically from the database?

Answer: Yes. Counts and latest registrations are fetched using SQL queries so the dashboard reflects current database contents.

2. Is access restricted to admin users?

Answer: The server checks the authenticated user's role before showing the dashboard or processing admin actions. Hiding a menu item alone is not security.

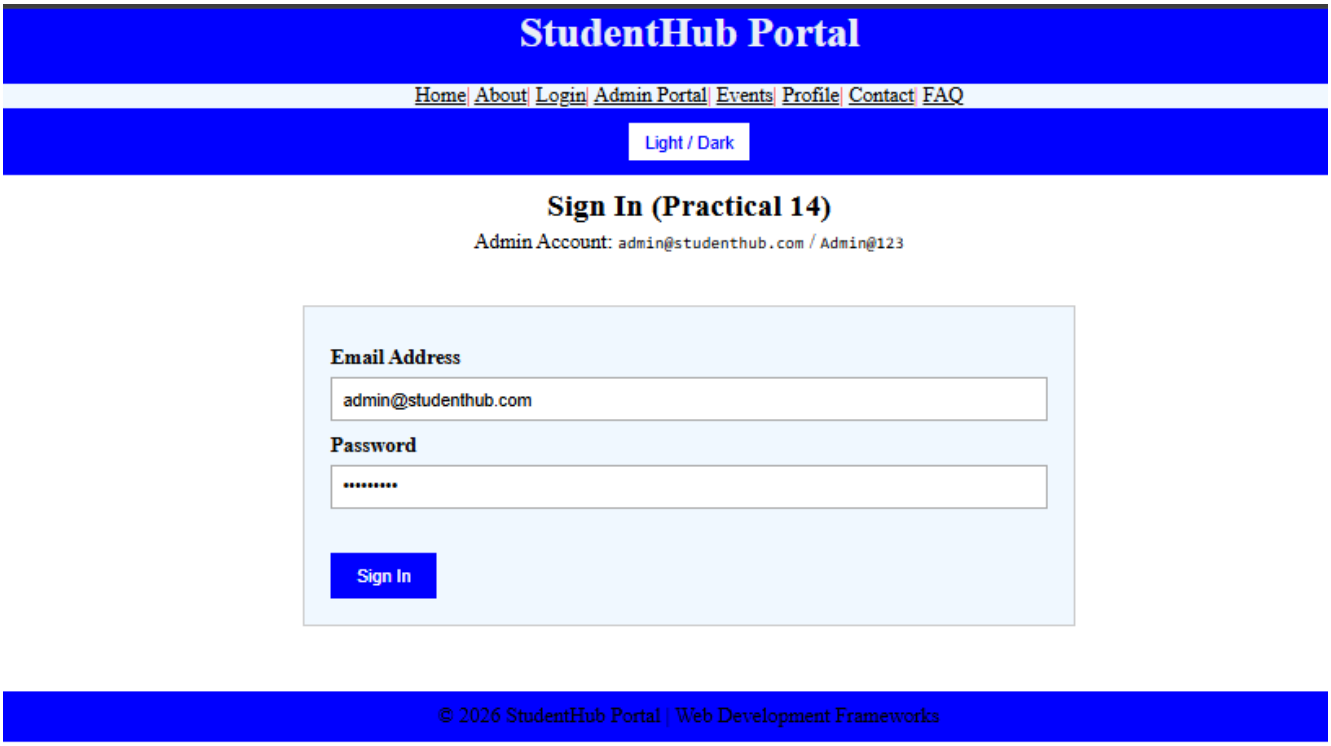
3. Are role-based menus working?

Answer: Menu links are generated according to the user's role, showing only the options that role is allowed to use.

4. Are important admin actions recorded in audit_logs?

Answer: Actions such as creating, updating, or deleting events and changing student records can be logged with the actor, action, target, and timestamp.

M. Evidence of Work (Screenshots)



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StudentHub Portal

Home | About | Admin Dashboard | Events | Profile | Logout (Admin: Admin User)

Light / Dark

Practical 14: Dynamic Admin Dashboard & Audit Log

Authenticated Admin: **admin@studenthub.com**

Total Students	Scheduled Events	Event Bookings	System Accounts
23	5	5	4
23 Active	Active Modules	Confirmed	RBAC Users

Student Status Toggle (Intermediate Extension)

ID	Name	Email	Course	Status	Action
23	nilesh bhatt	nil23@gmail.com	B.Tech Computer Engineering	Active	Set to Inactive
22	archi bhatt	25dce009@charusat.edu.in	B.Tech Computer Engineering	Active	Set to Inactive
21	vatsal makwana	25dce050@charusat.edu.in	B.Tech Computer Engineering	Active	Set to Inactive
20	Kavya Barot	kavya.barot@charusat.edu.in	B.Tech Information Technology	Active	Set to Inactive
19	Pranav Makwana	pranav.makwana@charusat.edu.in	B.Tech Mechanical Engineering	Active	Set to Inactive
18	Meera Solanki	meera.solanki@charusat.edu.in	B.Tech Civil Engineering	Active	Set to Inactive

Latest Event Registrations (SQL Aggregation / JOIN)

Booking #	Student Name	Event Title	Registered Date	Status
5	**Karan Desai**	Inter-College Sports Meet	2026-10-21 02:17:45	Confirmed
4	**Ananya Joshi**	AI & Machine Learning Seminar	2026-10-21 02:17:45	Confirmed
3	**Rohan Mehta**	Annual Hackathon 2026	2026-10-21 02:17:45	Confirmed
2	**Diya Shah**	Web Development Workshop	2026-10-21 02:17:45	Confirmed
1	**Aarav Patel**	Web Development Workshop	2026-10-21 02:17:45	Confirmed

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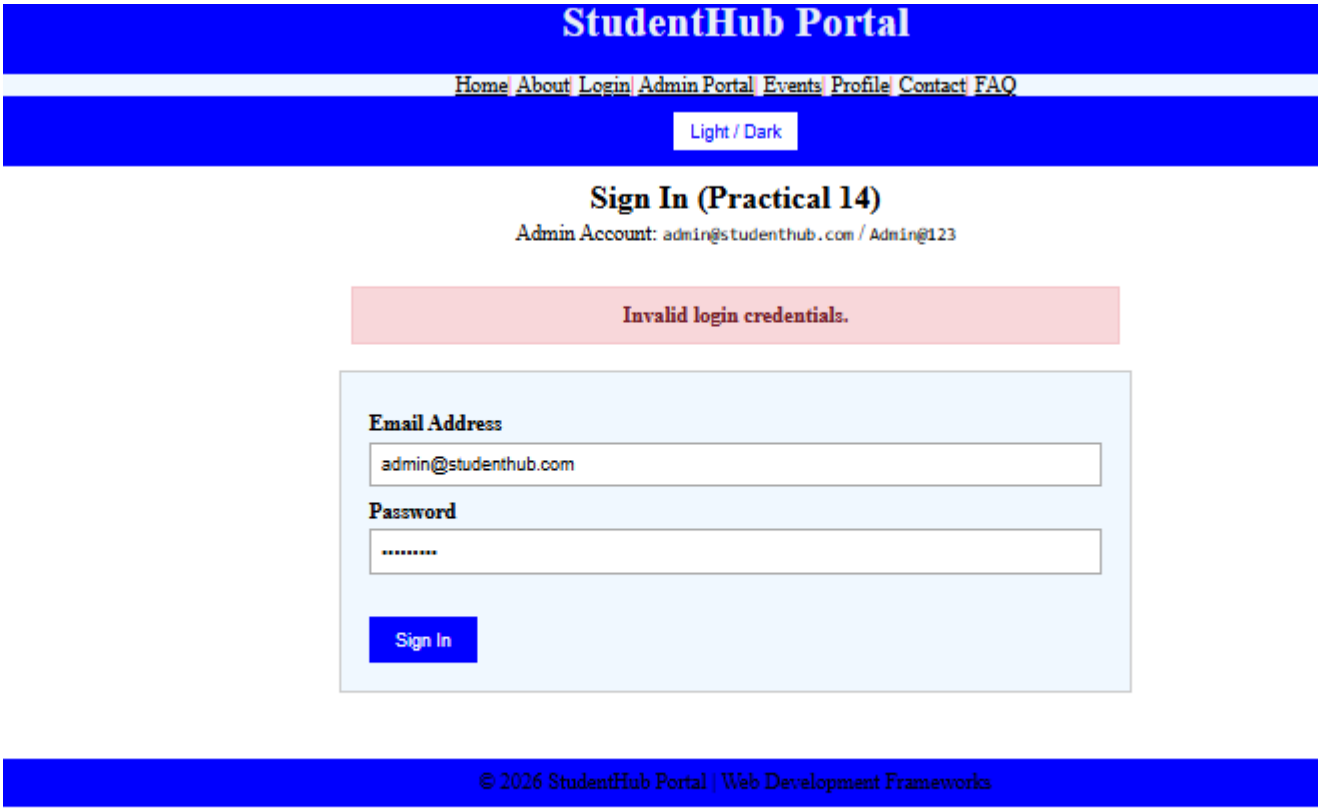
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1	**Aarav Patel**	Web Development Workshop	2026-10-21 02:17:45	**Confirmed**

Recent Audit Logs (Traceability & Security)

Log ID	User	Action	Details	IP Address	Timestamp
5	admin@studenthub.com	**ADMIN_LOGIN**	Admin logged into dashboard	::1	2026-10-21 03:15:23
4	student@studenthub.com	**USER_LOGIN**	Successful session login	::1	2026-10-21 02:47:02
3	admin@studenthub.com	**USER_LOGIN**	Successful session login	::1	2026-10-21 02:44:58
2	25dce009@charusat.edu.in	**USER_REGISTRATION**	New user registered via Practical 9	::1	2026-10-21 02:37:59
1	25dce050@charusat.edu.in	**USER_REGISTRATION**	New user registered via Practical 9	::1	2026-10-21 02:36:18

Logout Admin Session



N. GitHub Repository Link

https://github.com/vatsal-1950/StudentHUB.git